slopgs — MIDI Implementation Chart

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This document describes the MIDI behavior of the `slopgs` MIDI implementation, an emulation of the Microsoft GS Wavetable Synth.

1. Basic Specifications

Number of Parts

16

Maximum Polyphony

54 voices (48 sustained under continuous overload)

Effects

None

Wavetable source

A DLS-1 sample collection, usually `gm.dls`

Output format

22050 Hz, stereo, 16-bit signed PCM

2. Receive data

`n` is the MIDI channel nibble: `0` is channel 1, `F` is channel 16.

2.1. Channel Voice Message

`8n kk vv` — **Note Off**

- `kk` = note number (`00`–`7F`, 0–127)
- `vv` = release velocity. **Ignored.**

See 4.4.

`9n kk vv` — **Note On**

- `kk` = note number (`00`–`7F`, 0–127)
- `vv` = velocity (`01`–`7F`, 1–127)

See 4.3.

`An kk vv` — **Polyphonic Key Pressure**

Ignored.

`Bn cc vv` — **Control Change**

See 2.2.

`Cn pp` — **Program Change**

- `pp` = program number (`00`–`7F`, 0–127)

Selects the instrument from `pp` and the current Bank Select value (2.2).

`Dn vv` — **Channel Pressure**

Ignored.

`En ll mm` — **Pitch Bend Change**

- `ll` = LSB, `mm` = MSB
- Value = `mm` × 128 + `ll` (14-bit, 0–16383)
- Center = `00 40` (8192)

RPN 0 sets the range (2.2). See 4.5.

2.2. Control Change

The synthesizer recognizes only the controllers below. It ignores all other controller numbers. For the default values, see 3.4.

`Bn 00 mm` — **Bank Select MSB**

`Bn 20 ll` — **Bank Select LSB**

Sets the bank for the next Program Change. Notes that already sound do not change.

GS mode only. When GS mode (2.6.1) is off, the synthesizer ignores these messages and does not keep the value.

`Bn 01 vv` — **Modulation**
Recognized. See 4.7.

`Bn 06 mm` — **Data Entry MSB**
`Bn 26 ll` — **Data Entry LSB**
Sets the value of the selected RPN (see RPN, below). If RPN 0, 1 or 2 is not selected, these messages have no effect.

`Bn 07 vv` — **Channel Volume**
Recognized.

`Bn 0A vv` — **Pan**
Recognized. `00` = left, `40` (64) = center, `7F` = right. See 4.8.

`Bn 0B vv` — **Expression**
Recognized.

`Bn 40 vv` — **Hold 1 (Sustain)**
- `00` = off (pedal up)
- `01`–`7F` = on (pedal down)

`Bn 5B vv` — **Effects 1 Depth (Reverb Send)**
Ignored.

`Bn 5D vv` — **Effects 3 Depth (Chorus Send)**
Ignored.

`Bn 62 ll` — **NRPN LSB**
`Bn 63 mm` — **NRPN MSB**
Deselects the current RPN. These messages do not select a parameter.

`Bn 64 ll` — **RPN LSB**
`Bn 65 mm` — **RPN MSB**
Select the parameter that Data Entry (CC6/CC38) sets:

RPN (MSB LSB)	Parameter	Data Entry	Applied
`00 00`	Pitch Bend Sensitivity	MSB = semitones. Default 2. LSB is ignored.	Immediately, also to sounding notes.
`01 00`	Fine Tuning	MSB and LSB: −100 … +99 cents.	At the next Note On.
`02 00`	Coarse Tuning	MSB: −64 … +63 semitones. LSB is ignored.	At the next Note On. Melodic Parts only.
`7F 7F`	Null (no parameter)	No effect.	—
Other	—	Stored. No effect.	—

`Bn 78 00` — **All Sound Off**
Releases all sounding notes on the channel, also the notes that the pedal holds.

`Bn 79 vv` — **Reset All Controllers**
Resets Expression, Pitch Bend, Modulation and the sustain pedal. If `vv` is not `00`, it also resets Channel Volume and Pan. See 4.10.

`Bn 7A 00` — **Local Control**
Ignored.

`Bn 7B 00` — **All Notes Off**
Releases all sounding notes on the channel. Notes that the pedal holds continue until the pedal is up.

`Bn 7C 00` — **Omni Off**
`Bn 7D 00` — **Omni On**
Ignored (2.3).

`Bn 7E mm` — **Mono On**
Sets the channel to mono. Releases all sounding notes on the channel, also the notes that the pedal holds. `mm` is ignored.

In mono, each Note On first releases all notes on the channel, also the notes that the pedal holds. Notes do not glide.

`Bn 7F 00` — **Poly On**
Sets the channel to poly. Releases all sounding notes on the channel, also the notes that the pedal holds.

2.3. Channel Mode Message

The synthesizer always operates in **Omni Off**. It ignores Omni On and Omni Off.

CC126 and CC127 set Mono or Poly for each channel. The default is **Poly**.

2.4. System Common Message

Ignored: Song Position Pointer, Song Select, Tune Request.

`slopgs` plays a Standard MIDI File on its own clock. It does not use sync messages.

2.5. System Real Time Message

`F8` **Timing Clock,** `FA` **Start,** `FB` **Continue,** `FC` **Stop**
Ignored.

`FE` **Active Sensing**
Ignored.

`FF` **System Reset**
Sets the values in 3.4. Some values do not reset. See 4.10.

2.6. System Exclusive Message

The synthesizer recognizes the four messages below. It does not check the device ID (`dev`). It ignores all other System Exclusive messages.

`F0 41 dev 42 12 aa bb cc dd...dd sum F7`	Roland GS message (DT1). • `aa bb cc` = address (2.6.1) • `dd...dd` = data • `sum` = checksum. **Not checked.**
`F0 7E dev 09 01 F7`	GM System On. Resets the device and clears GS mode. Loads the instrument set.
`F0 7E dev 09 02 F7`	GM System Off. Resets the device and clears GS mode. **Unloads the instrument set**: no note sounds until GM System On or System Reset.
`F0 7F dev 04 01 ll mm F7`	Master Volume. Uses `mm` only. `ll` is ignored.

2.6.1. Roland Messages

Put these addresses in `aa bb cc` of the Roland GS message.

> **NOTE**
> When GS mode is off, the synthesizer ignores all addresses except GS Reset (`40 00 7F`).

Address	Parameter	Effect
40 00 7F	GS Reset	Resets the device and sets GS mode.
40 1n 02	RCV CHANNEL	Sets the MIDI channel that Part n receives. 01–10 = channel 1–16. 00 is ignored. Stays until the next reset.
40 1n 15	USE RHYTHM PART	00 = melodic Part. Other values = rhythm (drum) Part. Also resets all channels. See 4.10.
40 1n 40	Scale Tuning	12 values for Part n, C to B. 40 = no change. See 4.11.
40 1n *other*	Other Part parameters	**Ignored.**

The synthesizer ignores all other addresses. Examples:

- System Parameters (40 00 00–06: Master Tune, Master Volume, Master Key-Shift, Master Pan)
- Drum Setup Parameters (41 xx xx)
- Bulk Dump (48 xx xx, 49 xx xx)
- Display Data (model ID 45, 10 00 xx)

It also ignores RQ1 (data request) messages.

3. MIDI Implementation Chart

Basic Channel

The MIDI channel that each of the 16 Parts receives. Parts are numbered 0–15, as the n in the 40 1n xx addresses (2.6).

Mode

The MIDI Mode messages (Omni On/Off, Mono/Poly). slopgs always operates in Omni Off. You can set Mono or Poly for each channel.

Note Number / True Voice

The note numbers that the synthesizer accepts, and the note numbers that it plays. Note 60 is middle C.

Velocity

If the velocity of Note On and Note Off has an effect.

After Touch

Key□s = polyphonic (per-note) aftertouch. Ch□s = channel aftertouch.

Control Change

See 3.2 and 2.2.

Program Change

The program numbers that a Program Change can select. This is the message value: one less than the instrument number on most instrument lists.

System Exclusive

See 2.6.

System Common, System Real Time

Sync messages. Not used, except System Reset (FF).

3.1. Implementation Chart

Func	Transmit	Accept	Remarks
Basic Channel Default	X	1 – 16	Not channel = Part. See 3.3.
Basic Channel Changed	X	1 – 16, each	With RCV CHANNEL. GS mode only.
Mode Default	X	Mode 3	Omni Off, Poly.
Mode Messages	X	Mono, Poly (CC126/127)	Omni On/Off received, no effect.
Mode Altered	X	Mode 3, 4	
Note Number	X	0 – 127	Depends on the sample collection.
True Voice	X	0 – 127	Depends on the sample collection.
Pitch Bender	X	O	14-bit.
Control Change	X	see below	See 3.2.
Prog Change	X	O	
True #	X	0 – 127	
System Exclusive	X	O	• GM On/Off • GS Reset • RCV CHANNEL • USE RHYTHM PART • Master Volume • Scale Tuning • **Device ID and checksum unchecked**
System Common	X	X	
System Real Time	X	X	Except Reset — see Aux Messages, below.
Aux Messages			
Local ON/OFF	X	X	
All Notes OFF	X	O (CC123)	
Active Sense	X	X	
Reset	X	O (FF)	
Velocity			
Note ON	X	O	
Note OFF	X	X	
After Touch			
Key's	X	X	
Ch's	X	X	

3.2. Control Change

CC	Transmit	Accept	Remarks
Bank Select 0 / 32	X	cond.	No effect until a GS Reset has been received.
Mod Wheel 1	X	O	
Data Entry 6 / 38	X	O	Only meaningful with RPN, below.
Channel Volume 7	X	O	Default 100.
Pan 10	X	O	Default 64 (center).

CC	Transmit	Accept	Remarks
Expression 11	X	O	Default 127.
Hold 1 64	X	O	
Reverb Send 91	X	X	No reverb processor exists.
Chorus Send 93	X	X	No chorus processor exists.
NRPN 98 / 99	X	cond.	Only deselects the current RPN; no NRPN parameter is ever addressable.
RPN 100 / 101	X	O	Pitch Bend Sensitivity / Fine Tune / Coarse Tune only.
All Sound Off 120	X	O	Bypasses a held sustain pedal.
Reset All Ctrl 121	X	O	Bank/Program and RPN selection untouched.
Local Control 122	X	X	No local keyboard to disconnect.
All Notes Off 123	X	O	Honors a held sustain pedal.
Omni Off 124	X	X	
Omni On 125	X	X	
Mono On 126	X	O	Releases all notes, also the notes that the pedal holds.
Poly On 127	X	O	Releases all notes, also the notes that the pedal holds.
Other —	X	X	No effect, no error.

NOTE

- `O` : recognized/handled.
- `X` : received (or n/a) with no effect.
- `cond.` : recognized, but only under a stated condition — see Remarks.

Mode 1: Omni On, Poly.
Mode 2: Omni On, Mono.
Mode 3: Omni Off, Poly.
Mode 4: Omni Off, Mono.

3.3. Default Parts Mapping

At power-on, the Parts receive these channels:

Part	Channel	Assignment
0	10	Rhythm
1	1	Melodic
2	2	Melodic
3	3	Melodic
4	4	Melodic
5	5	Melodic
6	6	Melodic
7	7	Melodic
8	8	Melodic
9	9	Melodic
10	11	Melodic
11	12	Melodic
12	13	Melodic
13	14	Melodic
14	15	Melodic
15	16	Melodic

RCV CHANNEL and USE RHYTHM PART (2.6.1) change this map.

3.4. Default Parameters

Parameter	Value
Bank Select	0 / 0
Channel Volume	100
Pan	64 (center)
Expression	127
Pitch Bend	8192 (center)
Modulation	0
Sustain pedal	Released
Mono/Poly	Poly
Fine / Coarse Tuning (RPN1 / RPN2)	0 / 0
Data Entry	0
Master Volume	Unity
Per-Part tuning grid	Flat (0)
Part→channel map	Default (3.3)
Rhythm parts	Part `0` only
GS mode	Cleared (Bank Select inert until the next GS Reset)

4. Synthesizer Characteristics

4.1. Polyphony

The synthesizer has 54 voices. From the 49th voice, **it releases old voices faster than a steal-oldest policy does**.

A new note on a key that still sounds on the same channel **cuts the earlier note**. This also occurs when the earlier note is in its release.

The sample collection may let patches overlap. If so, both notes sound anyway.

A voice that is cut this way, or by an exclusive group in a drum kit, fades to silence in ~15ms or less. It does not click.

4.2. Envelopes

Levels fade at a **constant rate** in decibels.

The sample collection can **scale these times for each note**. Higher notes can decay faster. On some patches, harder notes attack faster.

4.3. Note On

Velocity behavior

Velocity uses a fixed curve: amplitude follows the square of velocity/127.

Velocity may also change the attack (e.g. a harder note attacks faster), if the sample collection specifies the patch this way.

CC behavior

In one instant, the synthesizer first collects all the Pitch Bend,

Volume, Expression, Pan and Modulation values. It then starts up the note with the values on or before the Note On.

If in the same instant those values change after the Note On, the note will glide to those new values.

4.4. Note Off

A released note fades at the rate of its patch's release, which does not change with the level at release. Therefore a note that is released at a lower level becomes silent sooner.

The synthesizer does not make short releases longer: a 10 ms release stops in 10 ms.

The fade can start up to approximately 25 ms before the Note Off (see 4.6).

4.5. Pitch Bend

Pitch Bend changes all sounding notes on the channel. This includes rhythm (drum) Parts, with the RPN 0 range of their channel.

RPN 0 accepts all values (`00`–`7F`). The bend itself stops at ±48 semitones.

4.6. Controller timing

Pitch Bend, Modulation, Channel Volume, Pan and Expression do not change in one step. Each change glides to the new value in approximately 25 ms or less (with `gm.dls`). The glide can start a short time before the message.

If a change and its reverse both occur in this time, you do not hear the change. Thus very fast Volume or Expression gates become soft. They never cut to silence.

4.7. Modulation

Modulation adds vibrato. Where the sample collection sets it, Modulation also adds tremolo (a change in loudness). The sample collection sets the depth.

Some patches have vibrato or tremolo when Modulation is `00`. In `gm.dls`, examples are Tremolo Str, Vibraphone, Shakuhachi, Flute and Contrabass.

Each patch sets its own rate and delay. If a patch does not set them, the rate is 5 Hz and there is no delay.

4.8. Pan

The pan law is constant power: each side is attenuated by `10·log10(position)`. At center, each side is approximately 3 dB down. At hard pan, the far side is approximately 25 dB down. It is not silent.

Each side has a maximum level (4.9). A loud Part can be at this maximum on both sides near center. Then Pan has no effect until the far side falls below the maximum. This range is wide at high Channel Volume and Expression, and narrow at low levels. At low levels, the full sweep is smooth. Near the maximum, the side that you pan toward does not become louder.

4.9. Loudness ceiling

Each voice has a maximum output level. A voice can reach it before velocity, Channel Volume and Expression are all at maximum. Above this level, a harder note is not louder. It can still have a different attack or sample layer. The patch's velocity sensitivity sets how wide this range is.

The ceiling also applies to the envelope. A note above the ceiling stays at full level until its envelope falls by the amount above the ceiling. Thus a loud note gets to full level sooner and starts to decay later. It sounds longer than a soft note.

4.10. Reset behavior

No reset changes:

- Pitch Bend Sensitivity (RPN 0)
- The selected RPN

Reset All Controllers (CC121) does not change Bank, Program, Fine Tuning (RPN 1) or Coarse Tuning (RPN 2).

GS Reset, GM System On, GM System Off and System Reset set Fine Tuning and Coarse Tuning to 0.

System Reset (`FF`) does not change the instrument of each Part. Only GS Reset and GM System On/Off set all Parts to program 0.

Exception: System Reset can discard a Program Change from up to approximately 46 ms before it.

USE RHYTHM PART (2.6.1) stops all notes on all channels immediately. It also resets Channel Volume, Pan, Expression, Pitch Bend and Modulation on all channels. Bank and Program do not change.

4.11. Scale Tuning

Each of the 12 values tunes one note name (C, C♯, D … B) by the value minus `40`, in cents: `00` = −64 cents, `40` = no change, `7F` = +63 cents. This also applies to rhythm Parts.

The synthesizer applies Scale Tuning at Note On, as for Fine Tuning. Sounding notes do not change.

The data always starts at C. If `F7` ends the message early, the synthesizer discards the last value before `F7`. It keeps the values before that one.

Other Part addresses (`40 1n` *other*) do not change Scale Tuning.

5. Presets

> **NOTE**
> All tables in this section depicts Microsoft's `gm.dls`, `slopgs` does not provide one. Your chosen DLS file will vary.

5.1. Instruments

#	Name (bank 0)	Variants (bank MSB: name)
	Piano	
0	Piano 1	8: Piano 1 16: Piano 1d
1	Piano 2	8: Piano 2
2	Piano 3	8: Piano 3
3	Honky-tonk	8: Honky-tonk
4	E.Piano 1	8: Detuned EP 1 16: E.Piano 1v 24: 60's E.Piano
5	E.Piano 2	8: Detuned EP 2 16: E.Piano 2v
6	Harpsichord	8: Coupled Hps. 16: Harpsichord 24: Harpsi.o
7	Clav.	—
	Chromatic Percussion	
8	Celesta	—
9	Glockenspiel	—
10	Music Box	—
11	Vibraphone	8: Vibraphone
12	Marimba	8: Marimba

#	Name (bank 0)	Variants (bank MSB: name)
13	Xylophone	—
14	Tubular-bell	8: Church Bell 9: Carillon
15	Santur	—
	Organ	
16	Organ 1	8: Detuned Or.1 16: 60's Organ 1 32: Organ 4
17	Organ 2	8: Detuned Or.2 32: Organ 5
18	Organ 3	—
19	Church Org.1	8: Church Org.2 16: Church Org.3
20	Reed Organ	—
21	Accordion Fr	8: Accordion It
22	Harmonica	—
23	Bandoneon	—
	Guitar	
24	Nylon-str.Gt	8: Ukulele 16: Nylon Gt.o 32: Nylon Gt.2
25	Steel-str.Gt	8: 12-str.Gt 16: Mandolin
26	Jazz Gt.	8: Hawaiian Gt.
27	Clean Gt.	8: Chorus Gt.
28	Muted Gt.	8: Funk Gt. 16: Funk Gt.2
29	Overdrive Gt	—
30	DistortionGt	8: Feedback Gt.
31	Gt.Harmonics	8: Gt. Feedback
	Bass	
32	Acoustic Bs.	—
33	Fingered Bs.	—
34	Picked Bs.	—
35	Fretless Bs.	—
36	Slap Bass 1	—
37	Slap Bass 2	—
38	Synth Bass 1	1: SynthBass101 8: Synth Bass 3
39	Synth Bass 2	8: Synth Bass 4 16: Rubber Bass
	Strings	
40	Violin	8: Slow Violin
41	Viola	—
42	Cello	—
43	Contrabass	—
44	Tremolo Str	—
45	PizzicatoStr	—
46	Harp	—
47	Timpani	—
	Ensemble	
48	Strings	8: Orchestra
49	Slow Strings	—

#	Name (bank 0)	Variants (bank MSB: name)
50	Syn.Strings1	8: Syn.Strings3
51	Syn.Strings2	—
52	Choir Aahs	32: Choir Aahs 2
53	Voice Oohs	—
54	SynVox	—
55	OrchestraHit	—
	Brass	
56	Trumpet	—
57	Trombone	1: Trombone 2
58	Tuba	—
59	MutedTrumpet	—
60	French Horns	1: Fr.Horn 2
61	Brass 1	8: Brass 2
62	Synth Brass1	8: Synth Brass3 16: AnalogBrass1
63	Synth Brass2	8: Synth Brass4 16: AnalogBrass2
	Reed	
64	Soprano Sax	—
65	Alto Sax	—
66	Tenor Sax	—
67	Baritone Sax	—
68	Oboe	—
69	English Horn	—
70	Bassoon	—
71	Clarinet	—
	Pipe	
72	Piccolo	—
73	Flute	—
74	Recorder	—
75	Pan Flute	—
76	Bottle Blow	—
77	Shakuhachi	—
78	Whistle	—
79	Ocarina	—
	Synth Lead	
80	Square Wave	1: Square 8: Sine Wave
81	Saw Wave	1: Saw 8: Doctor Solo
82	Syn.Calliope	—
83	Chiffer Lead	—
84	Charang	—
85	Solo Vox	—
86	5th Saw Wave	—
87	Bass & Lead	—
	Synth Pad	
88	Fantasia	—
89	Warm Pad	—
90	Polysynth	—
91	Space Voice	—

#	Name (bank 0)	Variants (bank MSB: name)
92	Bowed Glass	—
93	Metal Pad	—
94	Halo Pad	—
95	Sweep Pad	—
	Synth Effects	
96	Ice Rain	—
97	Soundtrack	—
98	Crystal	1: Syn Mallet
99	Atmosphere	—
100	Brightness	—
101	Goblin	—
102	Echo Drops	1: Echo Bell 2: Echo Pan
103	Star Theme	—
	Ethnic	
104	Sitar	1: Sitar 2
105	Banjo	—
106	Shamisen	—
107	Koto	8: Taisho Koto
108	Kalimba	—
109	Bagpipe	—
110	Fiddle	—
111	Shanai	—
	Percussive	
112	Tinkle Bell	—
113	Agogo	—
114	Steel Drums	—
115	Woodblock	8: Castanets
116	Taiko	8: Concert BD
117	Melo. Tom 1	8: Melo. Tom 2
118	Synth Drum	8: 808 Tom 9: Elec Perc.
119	Reverse Cym.	—
	Sound Effects	
120	Gt.FretNoise	1: Gt.Cut Noise 2: String Slap
121	Breath Noise	1: Fl.Key Click
122	Seashore	1: Rain 2: Thunder 3: Wind 4: Stream 5: Bubble
123	Bird	1: Dog 2: Horse-Gallop 3: Bird 2
124	Telephone 1	1: Telephone 2 2: DoorCreaking 3: Door 4: Scratch 5: Wind Chimes
125	Helicopter	1: Car-Engine 2: Car-Stop 3: Car-Pass 4: Car-Crash 5: Siren 6: Train 7: Jetplane 8: Starship 9: Burst Noise
126	Applause	1: Laughing 2: Screaming 3: Punch 4: Heart Beat 5: Footsteps
127	Gun Shot	1: Machine Gun 2: Lasergun 3: Explosion

5.2. Drums

Kits (use Program Change):

Program	Kit
0	Standard
8	Room
16	Power
24	Electronic
25	TR-808
32	Jazz
40	Brush
48	Orchestra
56	SFX

Notes, incl. GM+GS:

Note	Sound	Note	Sound
27	High Q	58	Vibraslap
28	Slap	59	Ride Cymbal 2
29	Scratch Push	60	Hi Bongo
30	Scratch Pull	61	Low Bongo
31	Sticks	62	Mute Hi Conga
32	Square Click	63	Open Hi Conga
33	Metronome Click	64	Low Conga
34	Metronome Bell	65	High Timbale
35	Acoustic Bass Drum	66	Low Timbale
36	Bass Drum 1	67	High Agogo
37	Side Stick	68	Low Agogo
38	Acoustic Snare	69	Cabasa
39	Hand Clap	70	Maracas
40	Electric Snare	71	Short Whistle
41	Low Floor Tom	72	Long Whistle
42	Closed Hi-Hat	73	Short Guiro
43	High Floor Tom	74	Long Guiro
44	Pedal Hi-Hat	75	Claves
45	Low Tom	76	Hi Wood Block
46	Open Hi-Hat	77	Low Wood Block
47	Low-Mid Tom	78	Mute Cuica

Note	Sound	Note	Sound
48	Hi-Mid Tom	79	Open Cuica
49	Crash Cymbal 1	80	Mute Triangle
50	High Tom	81	Open Triangle
51	Ride Cymbal 1	82	Shaker
52	Chinese Cymbal	83	Jingle Bell
53	Ride Bell	84	Bell Tree
54	Tambourine	85	Castanets
55	Splash Cymbal	86	Mute Surdo
56	Cowbell	87	Open Surdo
57	Crash Cymbal 2		